Customer Churn Modeling

RapidMiner coursework using a Kaggle dataset

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Understanding customer departure

The question

Which customer patterns were associated with churn, and how could analysis inform retention decisions?

The coursework

Used a Kaggle dataset in a machine-learning coursework project, recalled as a RapidMiner workflow.

Project scope

Prepared data, explored relevant features and evaluated predictive models to study churn patterns.

Evidence boundary

This presentation explains the approach. It does not report a verified model ranking, accuracy score or retention uplift.

Data preparation and feature analysis

01 Understand the churn label

Clarify what counts as leaving and when that outcome becomes known.

02 Prepare consistent inputs

Review missing values, data types and duplicates before model comparison.

03 Explore candidate predictors

Examine relationships with churn and exclude information revealed after departure.

A framework for comparing models

Baseline comparison

Compare candidate models with a simple baseline to determine whether added complexity improves prediction.

Held-out evaluation

Separate model development from evaluation. Keep preprocessing inside the training workflow to reduce leakage.

Useful metrics

Review precision, recall and F1 alongside class balance. Accuracy alone can hide missed churners.

Error review

Examine both missed churners and false alarms, then relate those errors to the cost of customer outreach.

Prediction should support a retention decision

Prioritize review

Use risk estimates to identify accounts for further review when outreach capacity is limited.

Explain the pattern

Connect model signals to understandable customer behavior and check whether the pattern is stable.

Test an intervention

Evaluate whether an outreach action changes outcomes before claiming that the model reduced churn.

Monitor over time

Track performance as customer behavior and product offerings change.

Analytical skills from the coursework

Data preparation
Feature analysis
Predictive model evaluation
Business interpretation

A customer analytics project focused on understanding churn through machine learning.

Dataset source: Kaggle
Tool recalled: RapidMiner
Context: academic coursework

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